

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
SIMATS ENGINEERING
COMPUTER SCIENCE AND ENGINEERING
CSA17 - ARTIFICIAL INTELLIGENCE

Experiment No. 5: Python Program for the Missionaries and Cannibals Problem

Aim

To write a Python program to solve the Missionaries and Cannibals problem, which involves transporting 3 missionaries and 3 cannibals across a river using a boat that can carry at most two people at a time, without ever leaving missionaries outnumbered by cannibals on either bank, using Breadth-First Search (BFS).

Algorithm

1. Start.
2. Represent a state as a tuple (M_left, C_left, Boat, M_right, C_right), where M and C denote the number of missionaries and cannibals on each bank, and Boat indicates which bank (1 = left, 0 = right) the boat is currently on.
3. Define the initial state as (3, 3, 1, 0, 0) — all missionaries, cannibals, and the boat on the left bank — and the goal state as (0, 0, 0, 3, 3).
4. Define a function `is_valid(state)` that checks: no count is negative, and on neither bank do cannibals outnumber missionaries (unless there are zero missionaries on that bank).
5. Define a function `get_successors(state)` that generates all possible next states by moving 1 missionary, 2 missionaries, 1 cannibal, 2 cannibals, or 1 missionary and 1 cannibal across the river in the direction the boat is currently on.
6. Filter out successor states that are invalid using `is_valid()`.
7. Use a queue to perform BFS: insert the initial state with an empty path, and mark it visited.
8. While the queue is not empty: a) Dequeue the front state along with its path. b) If this state equals the goal state, return the path as the solution. c) Otherwise, generate valid unvisited successors and enqueue them with the extended path.
9. If the queue is exhausted without reaching the goal state, report that no solution exists.
10. Print each intermediate state (bank counts and boat position) from the initial state to the goal state.
11. Stop.

Pseudocode

```
BEGIN
  start = (3, 3, 1, 0, 0)
  goal  = (0, 0, 0, 3, 3)
  CREATE queue Q, INSERT (start, path=[start])
  CREATE VISITED = { start }

  WHILE Q is not empty DO
    state, path = Q.dequeue()

    IF state == goal THEN
      PRINT "Solution found"
      PRINT path
      BREAK
```

```

    END IF

    FOR each successor IN get_successors(state) DO
        IF is_valid(successor) AND successor NOT IN VISITED THEN
            ADD successor TO VISITED
            Q.enqueue(successor, path + [successor])
        END IF
    END FOR
END WHILE

IF goal not reached THEN
    PRINT "No solution exists"
END IF
END

```

Result

The Python program for the Missionaries and Cannibals problem was executed successfully using Breadth-First Search. Starting from the state with all missionaries and cannibals on the left bank, the program explored valid boat-crossing moves while ensuring cannibals never outnumbered missionaries on either bank. A safe sequence of 11 moves was found that transported all missionaries and cannibals to the right bank, and each intermediate state was displayed. Thus, the Missionaries and Cannibals problem was solved successfully using Python.

Output Screenshot:

```

Terminal - missionaries.py

$ python3 missionaries.py
Initial State: 3 Missionaries, 3 Cannibals, Boat on Left Bank
Goal State:    All Missionaries & Cannibals moved to Right Bank

Solution found in 11 moves!

Step 0: Left Bank(M=3,C=3)  Right Bank(M=0,C=0)  Boat on Left
Step 1: Left Bank(M=3,C=1)  Right Bank(M=0,C=2)  Boat on Right
Step 2: Left Bank(M=3,C=2)  Right Bank(M=0,C=1)  Boat on Left
Step 3: Left Bank(M=3,C=0)  Right Bank(M=0,C=3)  Boat on Right
Step 4: Left Bank(M=3,C=1)  Right Bank(M=0,C=2)  Boat on Left
Step 5: Left Bank(M=1,C=1)  Right Bank(M=2,C=2)  Boat on Right
Step 6: Left Bank(M=2,C=2)  Right Bank(M=1,C=1)  Boat on Left
Step 7: Left Bank(M=0,C=2)  Right Bank(M=3,C=1)  Boat on Right
Step 8: Left Bank(M=0,C=3)  Right Bank(M=3,C=0)  Boat on Left
Step 9: Left Bank(M=0,C=1)  Right Bank(M=3,C=2)  Boat on Right
Step 10: Left Bank(M=1,C=1)  Right Bank(M=2,C=2)  Boat on Left
Step 11: Left Bank(M=0,C=0)  Right Bank(M=3,C=3)  Boat on Right

All missionaries and cannibals crossed safely!

```